

Pins become terminals

A global route starts and ends at GCell terminals, one tile per pin

The idea

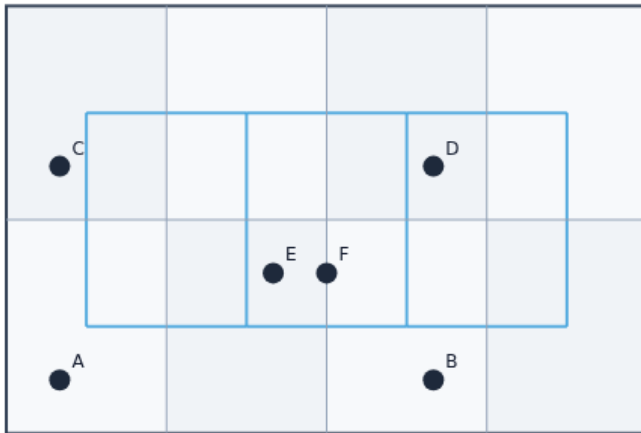
- Cell_gcell of x y returns column i equals floor of x over cell width clamped
- Cell A at one comma one maps to zero comma zero
- Cell D at eight comma five maps to two comma one
- Build a dictionary from cell id to GCell for every pin in the netlist

Pin → GCell

TERMINAL GCELLS

Pin → GCell

Step 1 / 5 · map · module01-03-terminal-gcells



Explanation

Cell centers map to GCells with $\text{floor}(x/\text{cellW})$, $\text{floor}(y/\text{cellH})$.

- Same as congestion grid
- i, j clamped

Terminals move with cells

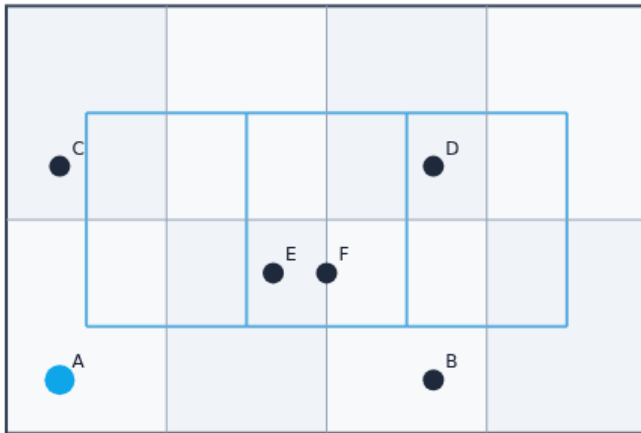
Pin → GCell

A → (0,0)

TERMINAL GCELLS

A → (0,0)

Step 2 / 5 · a00 · module01-03-terminal-gcells



Explanation

Spread A at (1,1) lands in GCell (0,0).

- Lower-left origin
- Clamp edges

A → (0,0)

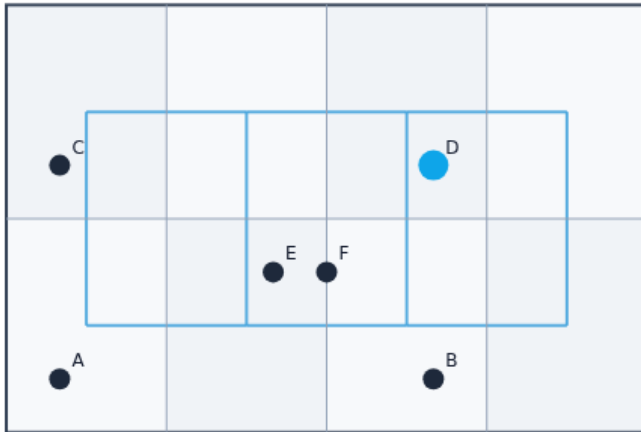
A → (0,0)

D → (2,1)

TERMINAL GCELLS

D → (2,1)

Step 3 / 5 · d21 · module01-03-terminal-gcells



Explanation

D at (8,5) → (2,1) top row.

- Top row $j=1$

D → (2,1)

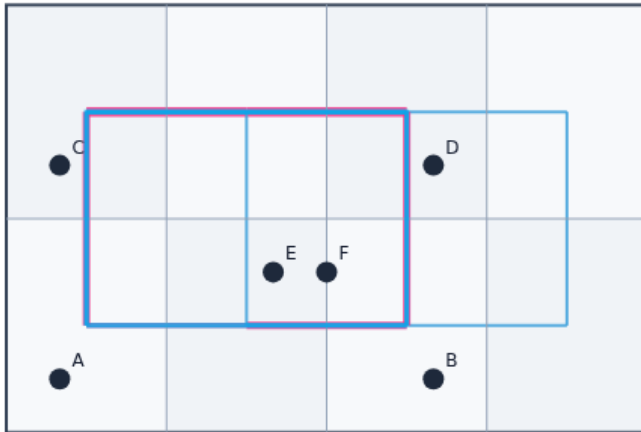
D → (2,1)

Routes on graph

TERMINAL GCELLS

Routes on graph

Step 4 / 5 · route · module01-03-terminal-gcells



Explanation

Global routes walk GCell edges between terminal tiles.

- Not bbox paint
- Edge usage

L-HV overlay

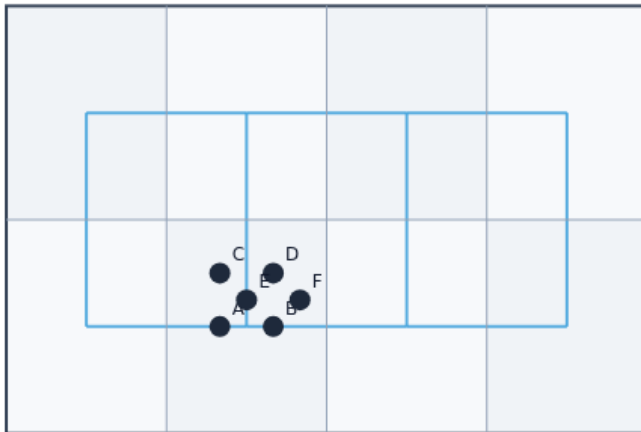
Routes on graph

Cluster seed

TERMINAL GCELLS

Cluster seed

Step 5 / 5 · cluster · module01-03-terminal-gcells



Explanation

Tight cluster maps many pins to one GCell—zero-length routes.

- Hot for study
- Spread for overflow

Same grid

Cluster seed

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Terminal GCell mapping

INTERACTIVE TOOL

Terminal GCell mapping

Map each cell center to the GCell tile that owns its pin.

Your job: on spread starter, A → (0,0) and D → (2,1). Move cells; terminals follow pins into GCells.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] A in (0,0)

Pick a challenge and click **Start**. Move cells, route, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓

Reset to starter

Reveal golden (study)

Route L-HV

Route L-VH

Route maze

Rip-up reroute

Graph

Reference / metrics

selected: A @ GCell (0,0)
routes: (none — click Route)
overflow total: 0.00 · max: 0.00 · count: 0
edge capacity: 2
HPWL: 34.0
view: your placement

Browser lab starter

learn_global_routing — 03 terminal gcells

Implement track

- Implement `terminals(positions, data)` returning the map
- Assert A through F on spread placement
- Print terminals before any routing lab

Pitfalls

- Using cell origin instead of pin center
- Off-by-one at the right or top chip edge without clamp
- Mixing site columns from legalization with GCell columns

Your turn

- Complete terminals for all six cells
- Next: L-shape pattern routes between two terminals